Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 4 to Draft 5

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Scope

Draft 5 (i) incorporates the two clarifications requested in ChatGPT's referee report on Draft 4 (recommendation: minor revision), and (ii) adds one new theorem: Target B holds for every irreducible representation of $D^\times$ trivial on K^3. Everything else is unchanged.

The two clarifications

1. Lemma `lem:centralizers`(a) is now stated for 2-elements y whose image in D_8 is nontrivial, with a parenthetical explaining that the central $\mu_2 \subset T$ of the odd case (whose centralizer is all of Γ_0) is excluded; the Brauer proof of Theorem `thm:reduction-all` only invokes the lemma for such y.

2. Lemma `lem:H1consequences`(c): the splitting $\tilde{V} \simeq \mu_3 \times V$ is now derived from $\tilde{V}$ being abelian *of exponent* 3 (cubes of lifts of K^3 are star lifts of elements of K^9, on which $\psi_6 = 1$; cubes of central lifts are star lifts of cubes, on which $\psi_6 = \theta_0^3 = 1$), with the remark that abelianness alone would not suffice ($C_9 \to C_3$).

Target B at level two (new Section 12.4)

For ρ irreducible, trivial on K^3, nonscalar on K^2: $\rho|_{K^1} = \chi_3 + \chi_3'$ (the two 3-dimensional characters of the Heisenberg group Q_1/Q_3), $\dim \rho = 6$, four such ρ per normalized central character. Findings, all by exact computation (scripts 13–15, about ten minutes):

1. The sufficient criterion of Proposition `prop:criterion` *fails* for these ρ: on the 219 good classes of $H = Q_3/Q_7$, F_ρ is a rational but not integral combination of the $\mathrm{Ad}(Q_1)$-invariant virtual characters of H (Smith normal form over $\mathbb{Z}$). For the three orbit sums of linear characters of the Heisenberg group it succeeds, consistent with Theorem `thm:levelone`.

2. Nevertheless $\Phi_\rho|_{\tilde{U}_3}$ is a genuine character. Since $\Phi_1|_{\tilde{U}_3} = \lambda_0 \cdot D$ with $D = \sum_{h \in \Lambda} \zeta^h$, $\Phi_\rho|_{\tilde{U}_3} = \lambda_0 \cdot (\Theta_\rho \circ r) D$, and the exact character table of H (297 classes, 243 linear and 54 three-dimensional characters) gives $((\chi_3 + \chi_3') \circ r) \cdot D = \sum\{\chi' \in \mathrm{Irr}(H) : \chi'(1) = 3\}$, each of the 54 exactly once.

3. Clifford theory then reduces Target B for ρ to the freeness of the $\mathrm{Ad}(Q_1)$-action on the 54 genuine three-dimensional irreducibles $\lambda_0\chi'$ of $\hat{H} = \tilde{U}_3/\ell(K^7)\ker\tilde{\theta}$ (a size-9 orbit would give a

non-integral multiplicity $e|O|/27$). Computing λ_0 on $\hat{H}$ in the cocycle model and the twisted action $\mu^t = \lambda_0 c_t(\chi' \circ \mathrm{Ad}\, t)$: exactly two orbits, both of size 27. Hence

$$\tau_\rho|_{\tilde{U}_1} = (\varepsilon_\rho \circ p) \otimes (\mathrm{Ind}(\lambda_0 \otimes \chi_1') \oplus \mathrm{Ind}(\lambda_0 \otimes \chi_2')),$$

two irreducibles of dimension 81, $\dim \tau_\rho = 162$ (Theorem `thm:leveltwo`, [P^c]). Cross-checks: the untwisted action on the same 54 characters has six orbits of size 9 (so the λ_0-twist is essential), and the 243 genuine linear characters form nine free orbits, recovering Theorem `thm:levelone`.

The cube-root correspondence (new Section 12.5, [O])

Both level theorems fit one pattern: to each irreducible χ of Q_1/Q_{m+1} corresponds an irreducible χ' of Q_3/Q_{3m+1} of the same degree with $\Phi_1 \cdot (\chi \circ r) = \Theta_{\mathrm{Ind}(\lambda_0 \otimes \chi')}$; this is formulated as a conjecture implying Target B for all ρ, and reduced to two statements: (i) $\mathrm{Ad}(Q_1)$ acts freely on the genuine irreducibles of $\hat{H}_m$ (equivalently every $\mathrm{Ind}_{\tilde{U}_3}^{\tilde{U}_1}\mu$ is irreducible), a statement about the cover alone; (ii) $(\chi \circ r)D$ is a virtual character of Q_3/Q_{3m+1}. Both hold for $m \leq 2$. For $m = 3$ the group has order 3^{11}; Burnside's algorithm as implemented will not reach it, but Clifford theory relative to Q_4/Q_{10} would.

Computational notes

`step13_level2.py` builds H from the Draft 1 model of Q/Q_7 (`finite.pkl`), computes its character table numerically (Burnside) and re-verifies it exactly in $\mathbb{Z}[\zeta_3]$ (all orthogonality relations), and solves the criterion over $\mathbb{Z}$ with a hand-written Smith normal form (`zsolve.py`). `step14_level2_restriction.py` decomposes $(\chi \circ r)D$. `step15_level2_clifford.py` computes λ_0 on $\hat{H}$ (formula (**??**) of the paper, as in script 11), the conjugation action with the exact cocycle, and the orbits; it also checks on 300 random pairs that λ_0 descends to a character of $\hat{H}$ (a consistency check, not load bearing). A scalar-extraction bug in a first version of script 15 (wrong coordinate for the F-component) was caught by that check.

Status

Theorem `thm:leveltwo` [P^c] (new); Theorems `thm:reduction-all`, `thm:H1-gen` [P] (clarifications applied); Section 12.5 [O]; nothing else changed.